LC Dynamics Simulator User Manual (Mac)

1. What the App Does

LC Dynamics Simulator calculates the time response of a liquid-crystal cell and displays transmittance, capacitance, applied voltage, internal orientation, potential, flow, polarization states, and viewing-angle characteristics.

The user controls the initial orientation, surface azimuth, twist angle, material constants, and voltage waveform. The app is therefore not limited to twisted-nematic (TN) operation: TN, STN, ECB (homogeneous), and VA (homeotropic, with $\Delta\varepsilon < 0$) cells can be studied, within these conditions:

- The electric field is along the cell thickness (z) only; in-plane switching such as IPS or FFS is not supported.
- Both substrates share one surface angle with strong anchoring; hybrid alignment with different angles is not supported.
- The polarizer is along x and the analyzer along y (crossed, fixed). Set the angle between polarizer and alignment with `Bottom azimuth` ϕ`0`.

This manual is an operation guide for first-time users. Equations, numerical methods, optical models, validation, and references are in the separate *LC Dynamics Simulator Technical Manual*.

Main functions include:

- Cell, material, drive, solver, and optical input panels
- Step off, Square wave, and arbitrary Waveform drive modes
- Accurate and Fast calculation modes
- Live progress, elapsed time, logs, and plots
- Time selection and animation after a run
- Viewing-angle transmittance and contrast maps
- CSV export and parameter Save/Load

2. Requirements, Installation, and Launch

2.1 Requirements

- macOS 14 (Sonoma) or later
- Tested on Apple silicon Macs
- Internet access is needed only to obtain and update the app through the Mac App Store

2.2 Install and Open

1. Download LC Dynamics Simulator from the Mac App Store.
2. Open Launchpad or the Applications folder in Finder.
3. Double-click `LC Dynamics Simulator`.

No separate Python installation, runtime, or command-line tool is required. Updates are distributed through the Mac App Store and do not delete parameter or CSV files saved by the user.

To launch with saved conditions, drop a parameter file on the app icon or choose the app from Finder's Open With menu.

2.3 Before the Mac App Store Release

From the development repository, open the built app:

```
open "mac/build/DerivedData/Build/Products/Release/LC Dynamics Simulator.app"
```

If it has not been built, open `mac/LCSimulator.xcodeproj` in Xcode and run it, or build the Release configuration as below. Xcode's Run uses the Debug configuration by default, which is more than ten times slower.

```
cd mac && xcodebuild -project LCSimulator.xcodeproj \
  -scheme LCSimulator -configuration Release \
  -derivedDataPath build/DerivedData build
```

3. Run Your First Calculation

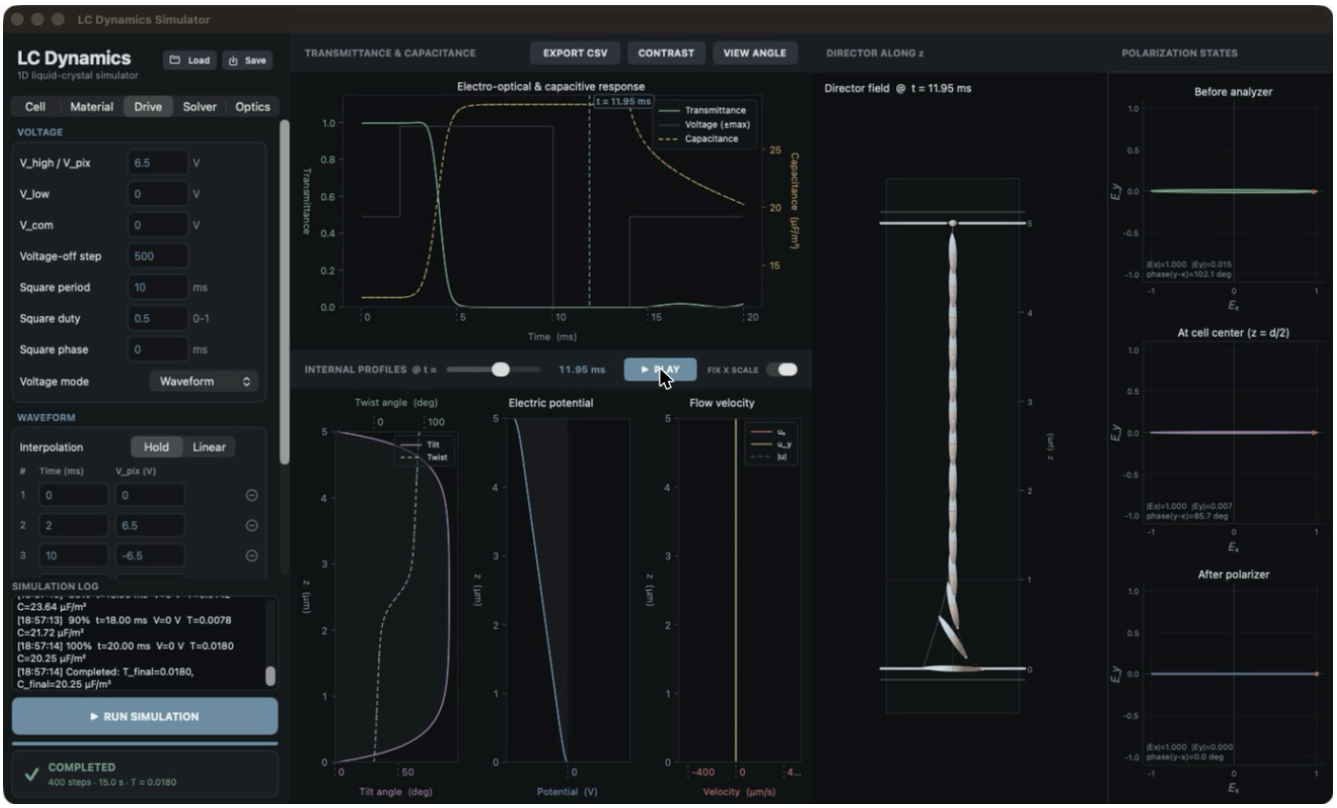
Use the defaults for the first run.

1. Open the app.
2. Leave the input values unchanged and press `RUN SIMULATION` at the lower left.
3. Wait until the status changes to `COMPLETED`.
4. Inspect transmittance, capacitance, and voltage in the top chart.
5. Move the time slider and observe the profiles, director display, and polarization display update together.
6. Press `PLAY` to animate the saved states; press `PAUSE` to stop playback.
7. Press `EXPORT CSV` to save the time response.

Press `STOP` to stop a run early. Wait until the run has stopped or completed before editing inputs for the next run.

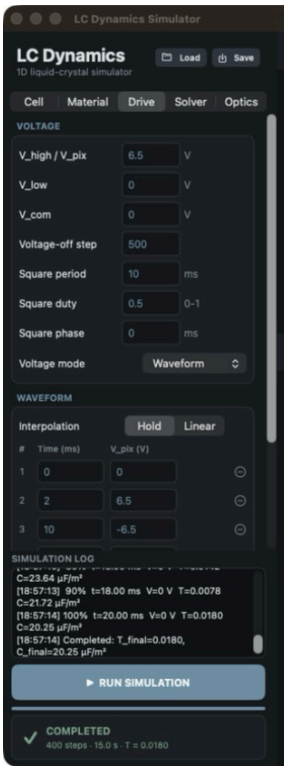
4. Window Guide

4.1 Main Window



Area	Purpose
Left	Inputs, parameter Load/Save, log, Run/Stop, progress
Center left	Time response, time slider, internal profiles
Center right	Director orientation through the cell
Right	Polarization at three positions

4.2 Input Panel



Tab	Main settings
`Cell`	Thickness, grid, surface angle, azimuth, twist, chiral pitch
`Material`	Elastic, dielectric, viscosity, and density values
`Drive`	Voltages, drive mode, square wave, arbitrary waveform
`Solver`	Time step, number of steps, calculation mode, saved-frame intervals
`Optics`	Refractive indices and wavelength

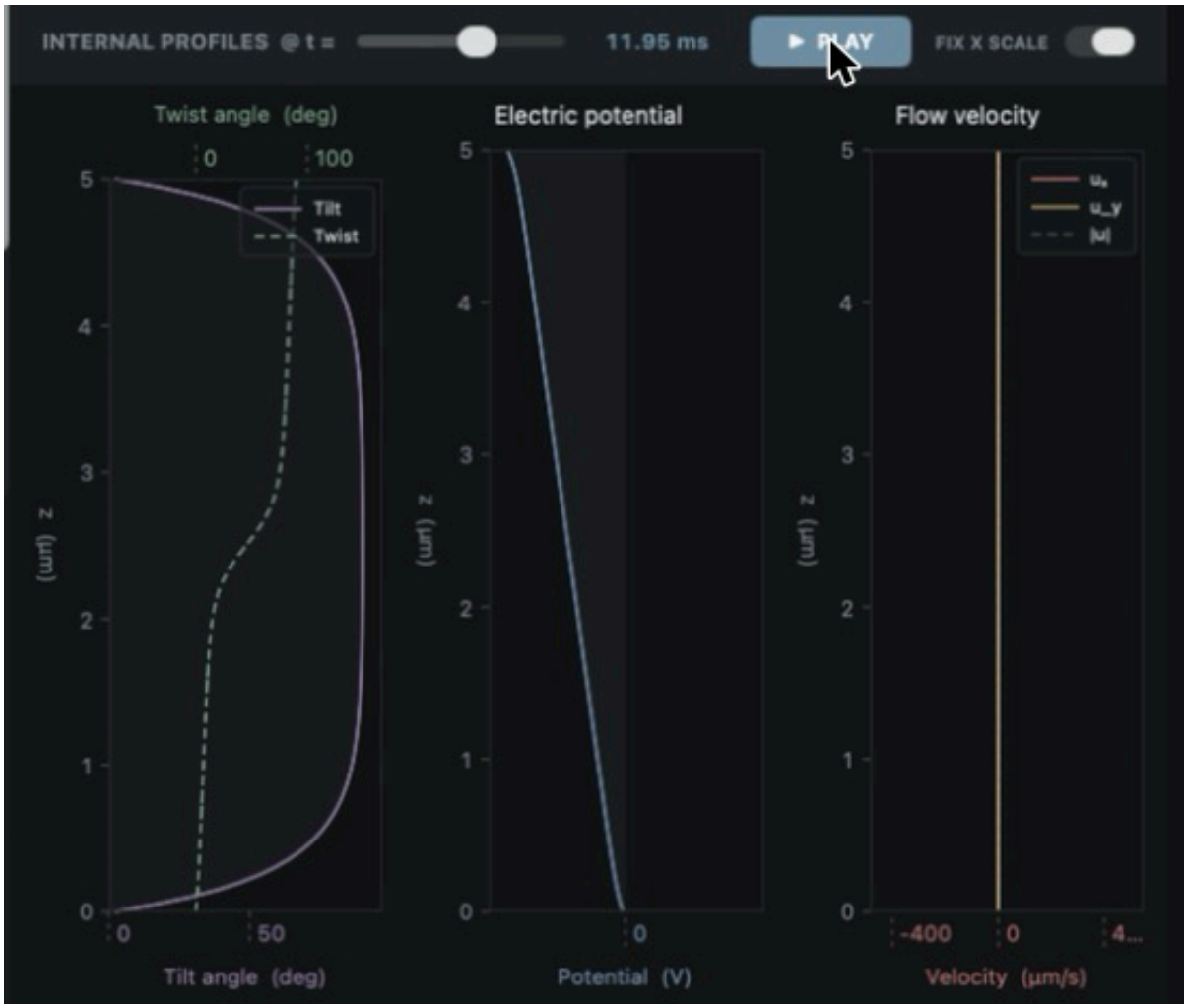
Click a field, replace its value, then press Return or click another field. Input that cannot be read as a number reverts to the previous value. Fields do not check value ranges, so verify that values are sensible. When a run starts, Grid points below 5 become 5, Total steps, Profile stride, and Live update stride below 1 become 1, and Square duty is limited to 0–1.

4.3 Time Response and Result Buttons

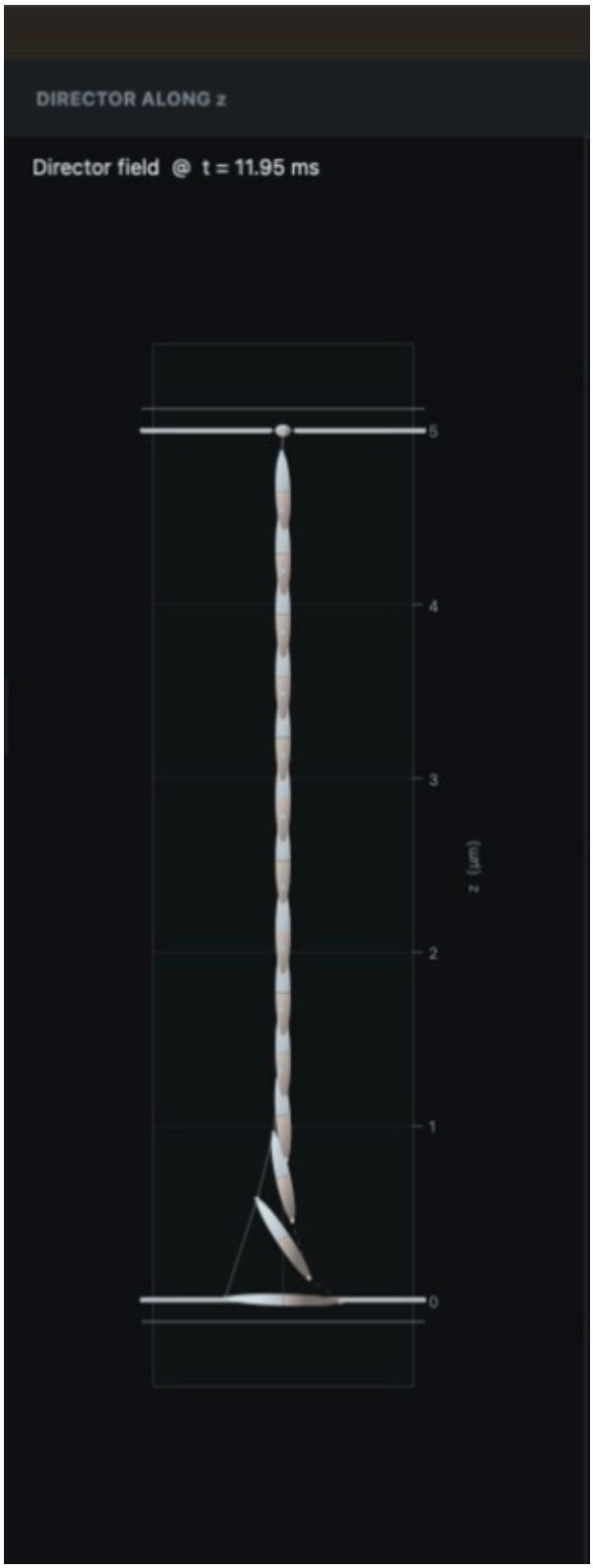


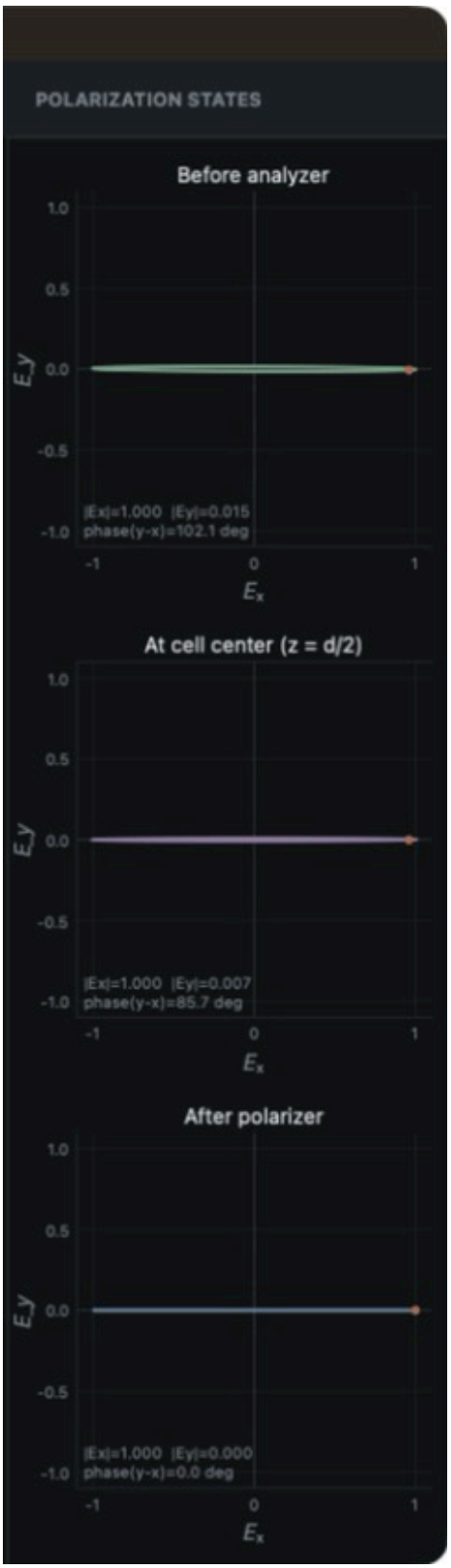
- Green: transmittance (left axis)
- Yellow dashed: capacitance (right axis)
- Thin blue-gray line: applied voltage, normalized as $0.98\,V/V_{\text{max}}$ on the transmittance axis. If the voltage includes negative values, $-V_{\text{max}}…V_{\text{max}}$ maps to 0…0.98 (0 V at height 0.49) and the legend reads `Voltage (±max)`
- Dotted brown line: voltage-off time in Step off mode (V off)
- Vertical dashed line: selected time
- `EXPORT CSV`: save the time response
- `VIEW ANGLE`: calculate transmittance versus viewing direction at one time
- `CONTRAST`: calculate viewing-angle contrast from two times

4.4 Profiles, Orientation, and Polarization



The three profile plots show orientation angles, electric potential, and flow at the selected time. Turn on `FIX X SCALE` to keep the horizontal ranges fixed during playback.





The rods show director orientation through the cell. They have no arrowheads because opposite director directions are physically equivalent. The three plots at right show polarization before the analyzer, at the cell center, and after the polarizer.

5. Enter Conditions

5.1 Cell Tab

Field	Meaning and use
Cell thickness	Cell gap in μm
Grid points	Number of points through the thickness; start with 101
Surface tilt θ	Polar angle of the surface director, measured from the substrate normal (z); shared by both substrates. 90° is parallel to the substrate, 0° perpendicular; the default 88° means about 2° pretilt
Bottom azimuth ϕ0	Azimuth at the lower substrate, measured from x (the polarizer); e.g. 45° for ECB
Twist angle	Azimuth difference between substrates; it is not restricted to 90°
Chiral pitch p	Chiral pitch in μm; 0 disables the chiral term

There is no liquid-crystal mode selector. Combine surface angles, azimuth, twist, pitch, and material values to represent the cell being studied.

`Surface tilt` θ is measured from the substrate normal, while `Tilt` in the profile chart is measured from the substrate plane, $\arcsin n_z$. They are related by Tilt $= 90^\circ - \theta$; with the defaults the surface `Tilt` is about 2°.

5.2 Material Tab

Enter elastic constants, dielectric constants, Leslie viscosity coefficients, and density. Units appear to the right of each field. Convert source data to the displayed units before entry, and check signs and decimal places carefully.

For a first study, run the defaults and then change one parameter at a time. This makes unexpected changes easier to trace.

5.3 Drive Tab

Step off Use Step off to study relaxation after a voltage change.

1. Choose `Step off`.
2. Enter `V_high / V_pix`, `V_low`, and `V_com`.
3. Enter the switching step in `Voltage-off step`.
4. Confirm the time step and total steps on the Solver tab.

Square wave

1. Choose `Square wave`.
2. Enter high and low voltages.
3. Enter `Square period` in ms.
4. Enter `Square duty` from 0 to 1.
5. Set `Square phase` when a shifted starting phase is needed.

Waveform Choose Waveform for pulses, ramps, polarity reversals, or multilevel drive. See section 6.

Field	How to use it

5.4 Solver Tab

Field	How to use it
Time step Δt	Smaller values improve time resolution but require more work
Total steps	Together with Time step, determines the total simulated time
Calculation mode	Use `Fast` while exploring and `Accurate` for final results
Profile stride	Smaller values save more frames and allow finer time selection
Live update stride	Larger values reduce live drawing overhead
Director relax	Director relaxation factor (default 0.5); reduce it if convergence is difficult
Voltage relax	Potential relaxation factor (default 0.8; used in Accurate mode)
Flow relax	Flow relaxation factor (default 0.8)

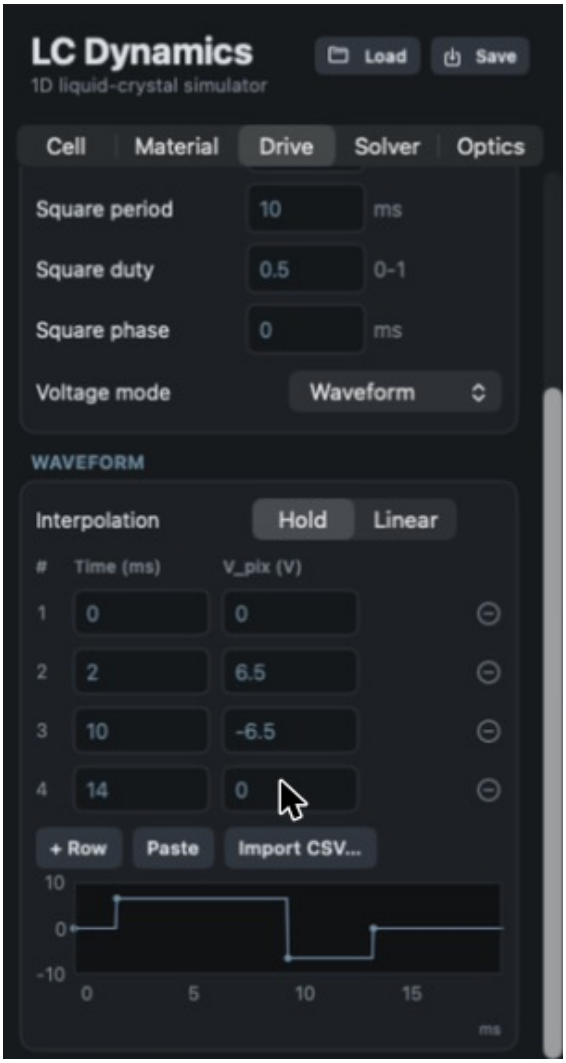
Start with the defaults. For a long job, first use Fast and fewer steps to check the setup, then repeat with Accurate.

5.5 Optics Tab

Enter extraordinary index `ne`, ordinary index `no`, and `Wavelength` λ. The same values are used for viewing-angle calculations.

6. Use Waveform Mode

6.1 Waveform Editor



Control	Action
`Hold`	Keep each row's voltage until the next row
`Linear`	Connect rows with straight-line ramps
`+ Row`	Add a row
⊖ at row right	Delete that row
`Paste`	Replace the table with two copied columns
`Import CSV…`	Replace the table with the first two columns of a text or CSV file

6.2 Enter a Waveform by Hand

1. Add the required rows.
2. Enter time in `Time (ms)`.
3. Enter voltage in `V_pix (V)`.
4. Choose Hold or Linear.
5. Confirm the waveform in the preview.

Rows are sorted by time for calculation. The first voltage is used before the first row, and the last voltage after the last row. An empty table produces 0 V.

6.3 Paste or Import Data

Prepare two columns: time in ms, then voltage in V.

```
time_ms,voltage_V
0,0
2,6.5
10,−6.5
14,0
```

Copy the columns from a spreadsheet and press `Paste`, or choose a file with `Import CSV…`. Both replace all existing rows. Commas, tabs, semicolons, and spaces are accepted. Headers and other nonnumeric lines are ignored. Always inspect the preview after import.

Common patterns include a Hold pulse, alternating positive and negative levels, a Linear ramp, and two different voltages at the same time for an instantaneous switch.

7. Run a Calculation

7.1 Pre-run Checklist

1. Confirm cell thickness and grid points.
2. Confirm material units, signs, and magnitudes.
3. Confirm drive mode and voltage values.
4. Confirm time step and total steps.
5. Check that `Time step × Total steps` covers the required time.
6. For Waveform, inspect the preview.
7. For final work, choose Accurate.

7.2 During a Run

Pressing `RUN SIMULATION` locks the inputs and changes the button to `STOP`. The status advances through preparation, initial equilibrium, and time evolution. Step-based progress, elapsed time, logs, and live charts appear after time evolution begins.

During initial equilibrium the progress bar sweeps rather than showing a percentage. This phase can take time for some settings; `Still working` does not necessarily indicate an error.

7.3 Stop or Quit

Press `STOP` to request cancellation. Initial-equilibrium work cannot stop inside its current operation, so CPU activity may continue briefly. Quitting during a run safely requests cancellation. Unsaved results are lost, so export needed data first.

8. Inspect Results

8.1 Select a Time

Move the slider below the time-response chart. The dashed line, profiles, director, and polarization update to the same saved time. `Profile stride` controls available times; use a smaller value and rerun when finer selection is required.

8.2 Play the Response

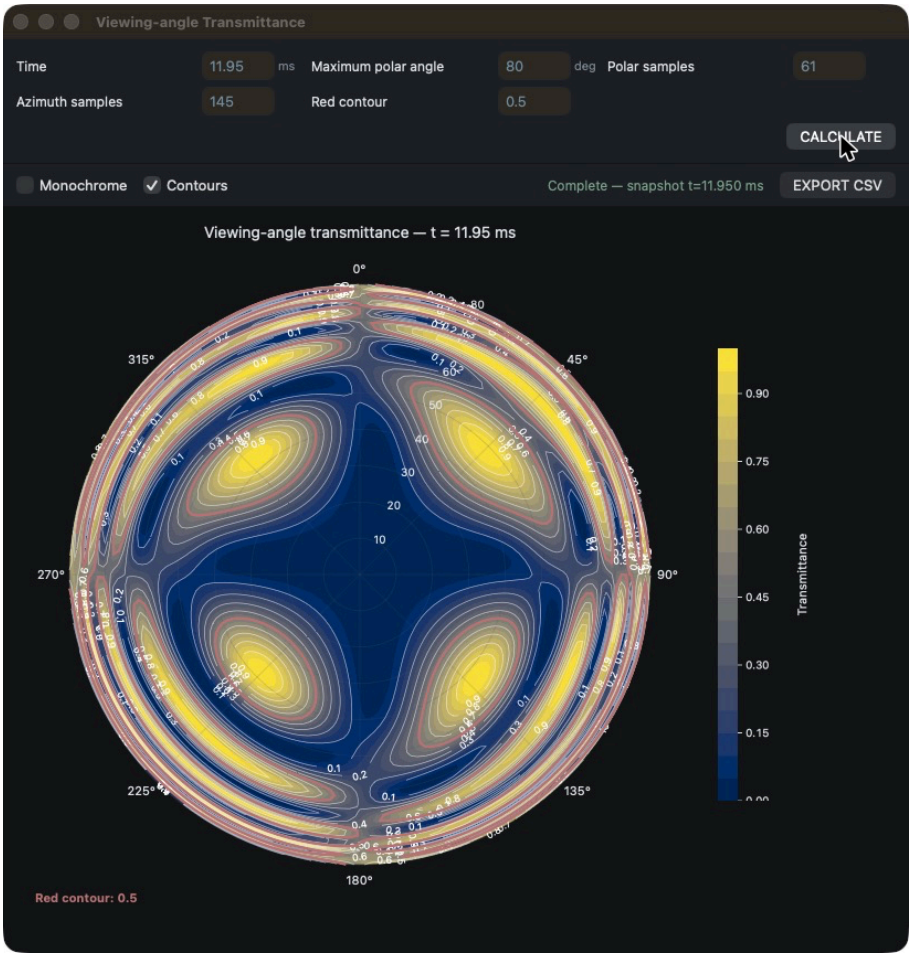
Press `PLAY`, use `PAUSE` to stop, and move the slider before pressing Play to restart from a chosen point.

8.3 Compare Curves

First compare transmittance with voltage timing, then inspect capacitance, orientation, potential, and flow. Turn on `FIX X SCALE` when changing axes would make time-to-time comparison difficult.

9. Viewing-Angle Maps

9.1 Transmittance



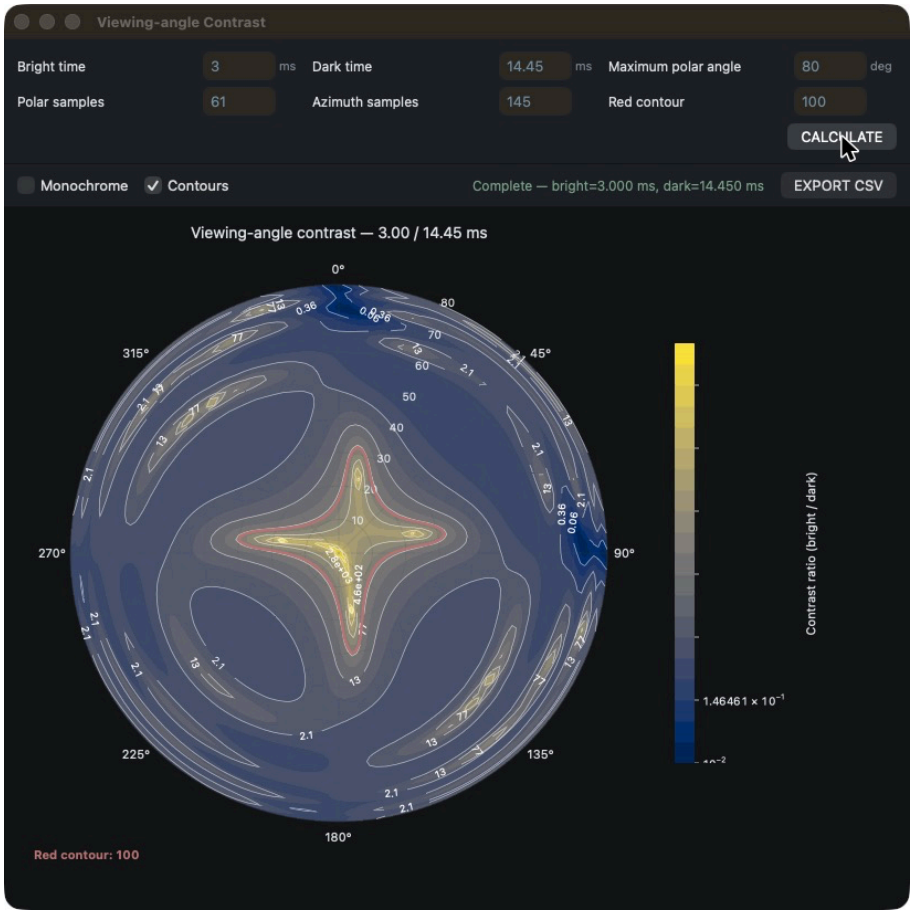
1. Complete the time-domain run.
2. Select a time with the slider.
3. Press `VIEW ANGLE`.
4. Enter the maximum polar angle and angular sample counts.
5. Optionally enter a transmittance value for the red contour.
6. Start the calculation and wait for completion.

7. Export CSV if needed.

The nearest saved frame is used; verify the actual time shown in the map window. More samples produce a smoother map but take longer.

The polar angle in the map is the propagation angle inside the liquid-crystal layer. Refraction from air into glass and liquid crystal is not modeled, so it differs from the viewing angle in air; convert with refraction in mind when comparing.

9.2 Contrast



1. Press CONTRAST.
2. Enter Bright time and Dark time.
3. Enter angular limits and sample counts.
4. Enter an optional highlighted contrast value.
5. Calculate and verify the actual times used.
6. Export CSV if needed.

Record both source times and conditions with a contrast result. Contrast can become very large where dark-state transmittance is extremely small.

10. Save Conditions and Results

10.1 Save and Open Parameters

Use Save or Save Parameters… (⌘S). The JSON file contains all inputs, the drive mode, and the waveform table, but no results. Use Load, Open Parameters… (⌘O), or drop the file on the app icon to restore it. Check drive mode, waveform, time step, and total steps after loading.

10.2 Export CSV

Use `EXPORT CSV` in the main window for the time response and in each map window for angular data. The main CSV contains step, time, voltage, capacitance, and transmittance. Map CSVs contain polar angle, azimuth, actual time, and transmittance or contrast. Files use Excel-compatible UTF-8 with a BOM.

11. Menus and Shortcuts

Action	Shortcut
Open Parameters	⌘O
Save Parameters	⌘S
Reset Parameters to Defaults	—
Export Results CSV	⌘E
Run / Stop	⌘R
Play / Pause	Space
Quit	⌘Q

12. Troubleshooting

Symptom	Action
No percentage after Run	Wait for initial equilibrium to finish and `Starting time evolution` to appear
Calculation is slow	Reduce grid points or steps, use Fast, or increase Live update stride
Memory use is high	Increase Profile stride to save fewer frames
CPU stays active after Stop	The current initial-equilibrium operation must return before stopping
Map buttons are disabled	Complete the time-domain run first
Map time differs from requested time	The nearest saved frame is used; reduce Profile stride and rerun
A value reverts	Check that it is a valid number and that integer-only fields have no decimals
Parameter file will not open	Stop the run and use JSON saved by this app
CSV will not save	Choose another writable location in the save dialog

Red lines in `SIMULATION LOG` summarize errors. For support, include the app version, macOS version, reproduction steps, parameter file, and log error. Find the version under About LC Dynamics Simulator.

13. Usage Notes

- The app uses a one-dimensional model through the cell thickness; the field is along z only, so in-plane switching is not modeled.
- Both substrates share one surface angle with strong anchoring, and the polarizer (x) and analyzer (y) are fixed and crossed.
- Viewing-angle polar angles are propagation angles inside the liquid crystal; interface refraction is not included.
- The director panel visualizes orientation; it is not a 3D spatial simulation.
- The user is responsible for checking units, signs, magnitudes, and model applicability.
- Increasing grid points, time steps, or angular samples increases run time.
- For final work, vary grid and time step to check numerical stability.
- Consult the separate Technical Manual when evaluating results for research or design.

14. Privacy and Support

The app requires no account and sends no personal data, simulation conditions, results, or usage information to the developer or third parties. It has no analytics, advertising, tracking, or cloud sync. All calculations run locally.

Parameter and CSV files are created only at locations chosen by the user. The app runs in the App Sandbox and accesses only explicitly selected files. Support contact details and the current privacy policy are available through the App Support and Privacy Policy links on the Mac App Store product page.